

Notes: Barber, Huang, Odean, and Schwarz (JF, 2022)

Attention-Induced Trading and Returns: Evidence from Robinhood Users

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1 Big picture

- **Question.** How do fintech brokerage design, simplified mobile trading, and investor attention affect retail trading behavior and stock returns?
- **Setting.** The paper studies Robinhood users from May 2018 to August 2020 using Robintrack stock-level user counts, CRSP returns and characteristics, TAQ retail trade measures, Google search volume, news coverage, and FINRA short-sale data.
- **Core empirical object.** Robintrack reports how many Robinhood users hold each security at frequent intraday snapshots. The authors use daily changes in the number of users holding a stock as a proxy for Robinhood user buying pressure.
- **Main conceptual idea.** Robinhood attracts many new and speculative investors and presents stocks through a simple app interface. This combination can make investors more likely to buy attention-grabbing stocks and to buy the same attention-grabbing stocks at the same time.
- **Main attention mechanism.** Retail investors face a search problem when deciding what to buy. Attention events, such as extreme returns, unusual volume, news, Google searches, earnings announcements, and prominent app lists, help determine which stocks enter the choice set.
- **Main behavior result.** Robinhood buying is much more concentrated than retail buying in TAQ. Around 35% of Robinhood net buying is concentrated in the top 10 stocks each day, compared with about 24% for the broader retail population.
- **App-design result.** Robinhood’s “Top Movers” list mixes large gainers and large losers based on absolute returns. Robinhood users buy both top gainers and top losers, whereas other retail investors buy top gainers much more aggressively than top losers.
- **Causal app-interface result.** Robinhood only includes stocks above a roughly \$300 million market-cap threshold in the Top Movers list. Around this discontinuity, otherwise similar top-mover stocks just above the threshold receive significantly more Robinhood user buying than matched stocks just below the threshold.
- **Outage result.** During unexpected Robinhood outages, retail trading falls disproportionately in popular Robinhood stocks and high-attention stocks. This supports the interpretation that Robinhood users are an important source of attention-induced retail trading.
- **Return result.** Intense Robinhood buying predicts sharp subsequent negative abnormal returns. The baseline top 0.5% daily Robinhood purchase-herding stocks lose about 4.7% over the next 20 trading days.
- **Intensity result.** The more extreme the Robinhood user-count increase, the more negative the subsequent return. For the most extreme herding threshold, subsequent 20-day abnormal returns approach -20% .
- **Market-participant response.** Abnormal short volume rises strongly around Robinhood herding events, suggesting that other traders recognize and trade against predictable Robinhood-induced price pressure.
- **Economic takeaway.** The paper does not argue that broad retail access is bad. Instead, it shows that simple, salient, and frictionless trading interfaces can coordinate attention, amplify

correlated buying, move prices temporarily, and hurt the investors who chase attention-driven episodes.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- Robinhood is an unusually clean setting for studying fintech-induced retail trading because its app makes trading simple, salient, and nearly frictionless.
- Robintrack provides a rare stock-day measure of aggregate Robinhood ownership. Although it does not observe individual trades or position sizes, daily changes in the number of users holding a stock reveal when many users newly enter the same stock.
- The sample covers a period in which Robinhood grew rapidly and retail trading surged during the COVID-19 period, generating substantial variation in attention and trading intensity.
- The setting also contains useful quasi-experimental variation: unexpected Robinhood outages and a sharp market-cap threshold for inclusion in the app’s Top Movers list.

Interpretation. The paper studies a platform whose design plausibly changes what investors notice. The empirical question is not simply whether retail investors trade a lot, but whether the interface focuses many users on the same stocks and whether that coordinated attention affects prices.

2.2 Main data sources

- **Robintrack:** stock-level snapshots of the number of Robinhood users holding a security from May 2, 2018 to August 13, 2020.
- **CRSP:** daily returns, prices, market capitalizations, and bid-ask spread information.
- **TAQ:** intraday trades, used to infer retail buys and sells through the Boehmer et al. fractional-penny retail-trade classification method.
- **Google Search Volume Index:** abnormal search intensity for tickers and firms.
- **Thomson Reuters MarketPsych Indices:** news article counts, used to construct abnormal news coverage.
- **FINRA short-sale data:** daily off-exchange short trading, used to test whether other traders trade against Robinhood herding events.

2.3 Robinhood user-count variables

The key raw variable is the number of Robinhood users holding stock i at time t . The paper converts this snapshot information into daily stock-level variables.

- **users_close**: the last observed Robinhood user count before the 4 p.m. close, conditional on the observation being after 2 p.m.
- **users_last**: the last observed user count on the day, regardless of timestamp.
- **userchg**:

$$userchg_{i,t} = users_close_{i,t} - users_close_{i,t-1}.$$

- **userratio**:

$$userratio_{i,t} = \frac{users_close_{i,t}}{users_close_{i,t-1}}.$$

Interpretation. The paper cannot observe Robinhood trade volume or dollars traded directly. It therefore treats a sharp increase in the number of users holding a stock as evidence that many users bought the stock during the day.

2.4 Robinhood herding events

The baseline herding event is:

$$rh_herd_{i,t} = 1$$

if stock i is in the **top 0.5%** of positive daily user-count percentage changes on day t , among stocks with at least **100 Robinhood users** on day $t - 1$.

This construction has three purposes:

- it focuses on positive purchase herding rather than ordinary ownership drift;
- it avoids tiny denominator cases by requiring at least 100 previous-day users;
- it identifies the most extreme daily episodes of coordinated Robinhood buying.

Interpretation. A herding event is not just a popular stock. It is a stock whose Robinhood ownership suddenly jumps by an unusually large percentage.

2.5 TAQ retail trading measures

The paper compares Robinhood behavior to broader retail trading inferred from TAQ.

- Retail buys and sells are classified using the fractional-penny price-improvement algorithm.
- **net_buys** equals retail buys minus retail sells.
- **taq_retimb** is a retail order imbalance measure:

$$taq_retimb_{i,t} = \frac{retail\ buys_{i,t} - retail\ sells_{i,t}}{retail\ buys_{i,t} + retail\ sells_{i,t}}.$$

Interpretation. TAQ is used as a benchmark for the broader retail-investor population. This lets the paper ask whether Robinhood users are simply retail investors in general or whether they trade in a more concentrated, attention-sensitive way.

2.6 Attention proxies

The paper studies whether herding is predictable from attention-related variables. Key attention proxies include:

- lagged Robinhood herding;
- whether the stock had a top-20 absolute return the prior day;
- abnormal trading volume;
- prior Robinhood user change and lagged Robinhood popularity;
- abnormal Google search volume;
- abnormal news coverage;
- earnings announcement indicators.

Abnormal volume, search, and news measures are generally log ratios of the current value to the trailing average over days $t - 21$ to $t - 2$.

2.7 Top Movers list and app-interface variables

A central app-design feature is Robinhood’s **Top Movers** list.

- It displays stocks with the largest absolute percentage price changes.
- Unlike many financial websites, it mixes large gainers and large losers in one list.
- The list contains only a small number of stocks and is salient inside the Robinhood interface.
- Robinhood requires stocks to have market capitalization above roughly \$300 million to appear on the list.

Interpretation. The Top Movers list is useful for identification because it creates a clear prediction: if the app matters, Robinhood users should respond strongly to both extreme gainers and extreme losers, and stocks just above the \$300 million cutoff should receive extra Robinhood buying.

2.8 Return and performance measures

The paper studies abnormal returns after herding events.

- Daily abnormal return is the stock return minus the CRSP value-weighted market return.
- Buy-and-hold abnormal returns compare the compounded stock return to the compounded market return.
- Calendar-time portfolio tests form portfolios of herding stocks and estimate alphas using CAPM, Fama-French, and momentum-factor models.
- The paper also computes aggregate Robinhood investor experience by comparing average purchase prices and average sale or end-of-window prices around events.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline attention-prediction model

The first model asks whether attention variables predict Robinhood purchase herding:

$$\begin{aligned} rh_herd_{i,t} = & \alpha + \beta_1 rh_herd_{i,t-1} + \beta_2 ExtremeAbsRet_{i,t-1} + \beta_3 AbnVol_{i,t-1} \\ & + \beta_4 UserChange_{i,t-1} + \beta_5 \ln(Users_{i,t-1}) + \beta_6 AbnSVI_{i,t-1} \\ & + \beta_7 AbnNews_{i,t-1} + \beta_8 EarningsAnnouncement_{i,t-1} + \varepsilon_{i,t}. \end{aligned}$$

- The dependent variable is an indicator for a Robinhood herding event.
- The model is estimated as a linear probability model.
- Standard errors are clustered by stock and day.

Interpretation. If Robinhood herding is attention-induced, attention proxies should predict which stocks experience a sudden surge in Robinhood ownership.

3.2 Concentration of buying and selling

For each day, the paper ranks stocks by positive Robinhood user changes and by positive TAQ net retail buying. It then computes the share of total net buying accounted for by the top-ranked stocks and a Herfindahl-Hirschman concentration index:

$$HH_t = \sum_i s_{i,t}^2,$$

where $s_{i,t}$ is stock i 's share of total positive net buying or selling on day t .

Interpretation. Higher concentration means many investors are acting on the same stock ideas at the same time. If Robinhood users are especially attention-driven, their buying should be more concentrated than general retail buying.

3.3 Robinhood outage regressions

The paper exploits unexpected Robinhood outages as shocks to Robinhood users' ability to trade. For the full-day March 2 outage, the specification is:

$$RetailPerc_{i,t} = a + bOutage_t + \mu_{tod} + \mu_{stock} + \varepsilon_{i,t}.$$

For the March 3 intraday outage, the specification is:

$$RetailPerc_{i,t} = a + bOutage_t + cRepair_t + \mu_t + \mu_{tod} + \mu_{stock} + \varepsilon_{i,t}.$$

For the June 18 intraday outage, the specification is:

$$RetailPerc_{i,t} = a + bOutage_t + cPartial_t + dRepair_t + \mu_t + \mu_{tod} + \mu_{stock} + \varepsilon_{i,t}.$$

- $RetailPerc_{i,t}$ is the percent of TAQ trades that are classified as retail for stock i in interval t .
- $Outage_t$ identifies the outage interval.
- $Repair_t$ identifies the hour or interval after systems are restored.
- $Partial_t$ identifies partial restoration for the June outage.
- The regressions include time-of-day and stock fixed effects, and sometimes day fixed effects.

Interpretation. If Robinhood users matter for attention-induced trading, retail trading should fall during outages, especially in stocks popular with Robinhood users or stocks predicted to attract Robinhood herding.

3.4 Top Movers ranking regression

To test how app display affects buying, the paper estimates:

$$NetBuy_{i,t} = \beta_0 + \beta_1 Score_{i,t} + \beta_2 \mathbf{1}\{R_{i,t} < 0\} + \beta_3 Score_{i,t} \times \mathbf{1}\{R_{i,t} < 0\} + \alpha_t + \varepsilon_{i,t}.$$

- $NetBuy_{i,t}$ is either Robinhood intraday user change or TAQ intraday net retail buying.
- $Score_{i,t}$ assigns 20 to the highest absolute-return top mover and 1 to the 20th highest.
- $\mathbf{1}\{R_{i,t} < 0\}$ indicates a top loser rather than a top gainer.
- Day fixed effects absorb market-wide daily conditions.

Interpretation. The key comparison is between Robinhood and TAQ. If Robinhood’s mixed Top Movers list matters, Robinhood users should buy both top gainers and top losers, whereas broader retail traders should respond more strongly to top gainers.

3.5 Regression discontinuity around the \$300 million threshold

The market-cap threshold specification can be summarized as:

$$RHChange_{i,t} = \gamma_0 + \tau \mathbf{1}\{MktCap_{i,t} > 300m\} + f_N(MktCap_{i,t} - 300m) + \varepsilon_{i,t},$$

where $f_N(\cdot)$ is a polynomial in distance from the threshold.

- The treatment group contains top-mover stocks just above \$300 million in market cap.
- The control group contains matched stocks just below \$300 million with similar absolute overnight returns.
- The coefficient τ measures the discontinuous increase in Robinhood buying for stocks eligible for the Top Movers list.

Interpretation. This design is much closer to causal evidence on app display: stocks just above and below the cutoff are similar in size and return movement, but only the above-cutoff stocks are eligible for display.

3.6 Event-time abnormal returns

The paper studies returns around herding event day 0. The buy-and-hold abnormal return is:

$$BHAR_{i,\tau} = \prod_{t=\tau-10}^{\tau} (1 + R_{i,t}) - \prod_{t=\tau-10}^{\tau} (1 + R_{m,t}).$$

For postevent returns, the same logic compounds stock returns and market returns over days after the event.

Interpretation. BHARs directly describe the return an investor would experience from holding the stock. They are more appropriate than simply summing daily abnormal returns when bid-ask bounce or temporary price pressure creates return reversals.

3.7 Calendar-time trading strategy

The calendar-time portfolio buys each herding stock at the close of the event day, holds for five days, and estimates:

$$R_{p,t} - R_{f,t} = \alpha + \beta(R_{m,t} - R_{f,t}) + \sum_{k=1}^K c_k F_{k,t} + \varepsilon_{p,t}.$$

The factor sets include CAPM, four-factor, and Fama-French five-factor plus momentum specifications.

Interpretation. This test addresses event-time cross-sectional dependence and overlapping-return concerns. A negative alpha means that a tradable strategy of buying Robinhood herding stocks performs poorly even after standard factor adjustments.

3.8 Daily return regressions with lags

The paper also estimates daily stock-return regressions of the form:

$$R_{i,t} = a + \sum_{k=1}^5 b_k rh_herd_{i,t-k} + \sum_{k=1}^5 c_k taq_retimb_{i,t-k} + \sum_{k=1}^5 d_k R_{i,t-k} + \mu_t + \varepsilon_{i,t}.$$

- The sum of the b_k coefficients gives the five-day abnormal return associated with Robinhood herding.
- Lagged returns help separate Robinhood herding from ordinary short-term reversal.
- TAQ retail order imbalance controls for broader retail trading pressure.
- Day fixed effects remove market-wide return shocks.

Interpretation. These regressions ask whether herding has predictive power for future returns after controlling for price reversal and general retail order imbalance.

3.9 Aggregate Robinhood investor experience

To evaluate whether the Robinhood community profits from herding events, the paper computes a weighted average purchase price and sale price. Conceptually:

$$PrcRatio_j = \frac{SPrc_j}{PPrc_j} - 1,$$

where $PPrc_j$ is the average purchase price and $SPrc_j$ is the average sale or end-of-window price for event j .

The paper also compares this with a counterfactual in which the same capital is traded in an S&P 500 ETF.

Interpretation. This converts stock-level reversals into an investor-outcome measure. It asks whether the community of Robinhood investors who trade during herding episodes buys too high and exits too low.

3.10 Short-trading response

The final mechanism test uses abnormal short volume:

$$AbShort_{i,t} = \alpha + \beta RHSignal_{i,t} + \Gamma Controls_{i,t} + \varepsilon_{i,t},$$

where $RHSignal_{i,t}$ is either the Robinhood user-change ratio or the Robinhood herding indicator.

Controls include returns, return-rank indicators, and abnormal news coverage. The paper reports Fama-MacBeth standard errors.

Interpretation. If sophisticated traders understand that Robinhood herding predicts reversals, abnormal short volume should increase precisely when Robinhood buying is extreme.

4 Identification and instruments (what makes the estimates credible)

4.1 Main identification problem

The core difficulty is separating Robinhood-driven attention trading from ordinary stock-level events. A stock that has a huge return, high volume, and heavy news coverage may attract all retail investors, not only Robinhood users. It may also have fundamental news that independently predicts future returns.

The paper addresses this problem by combining several complementary designs rather than relying on a single instrument.

4.2 Unexpected outages as trading-access shocks

The Robinhood outage tests are valuable because they temporarily prevent Robinhood users from trading while leaving other investors' brokerage access unchanged.

- The outages occur at specific times and are plausibly unrelated to individual stock fundamentals.
- Retail trading drops during outages.
- The drop is stronger in popular Robinhood stocks and high-attention stocks.
- Trading rebounds when service is restored.

Identification interpretation. Outages help show that Robinhood users are not just passively correlated with retail trading. Their inability to trade measurably changes retail trading patterns.

4.3 Top Movers market-cap discontinuity

The \$300 million Top Movers cutoff creates a sharp eligibility rule.

- Stocks just above and just below \$300 million are relatively similar.
- Both groups can have similar extreme absolute overnight returns.
- Only stocks above the threshold can appear on Robinhood's Top Movers list.
- A discontinuous jump in Robinhood buying above the threshold is therefore strong evidence that display affects trading.

Identification interpretation. The RD design is the paper's cleanest evidence that app interface, not only investor composition, influences Robinhood user trading.

4.4 Comparison with broader retail investors

The paper repeatedly compares Robinhood user changes with TAQ-implied retail trading.

- Robinhood buying is more concentrated than TAQ retail buying.
- Robinhood users buy both top gainers and top losers; TAQ retail buyers tilt more toward gainers.
- Robinhood herding predicts returns more strongly than TAQ herding alone.

Identification interpretation. These comparisons show that the paper is not simply rediscovering generic retail trading. Robinhood users display a distinctive attention pattern.

4.5 Return-predictability controls

The return tests address several alternative explanations.

- Lagged returns control for ordinary short-term reversal.
- TAQ retail order imbalance controls for broader retail buying pressure.
- Quote-midpoint tests address bid-ask bounce.
- Subsample tests show stronger effects in small-cap stocks and during the COVID retail-trading surge, where retail price pressure should matter more.

- Overnight-return splits show that negative future returns occur even after herding that follows negative overnight returns, not only after positive-return attention shocks.

Identification interpretation. The negative returns are not just a mechanical reversal after prior gains. They are tied to extreme Robinhood buying pressure.

4.6 What the paper can and cannot prove

- The paper can show that Robinhood user herding is attention-related, partly app-induced, and strongly predictive of negative returns.
- The paper can show that Robinhood users are an unusually strong proxy for attention-induced retail trading.
- The paper cannot observe individual Robinhood accounts, trade sizes, wealth, demographics, or exact dollar flows.
- The paper cannot fully separate trades by Robinhood users from trades by other attention-driven investors who notice the same stocks.
- The paper does not claim that all Robinhood users lose money or that all fintech access is harmful. The strongest conclusion concerns extreme attention-induced herding episodes.

5 Tables and figures

5.1 Figure 1: Robinhood and TAQ retail trading

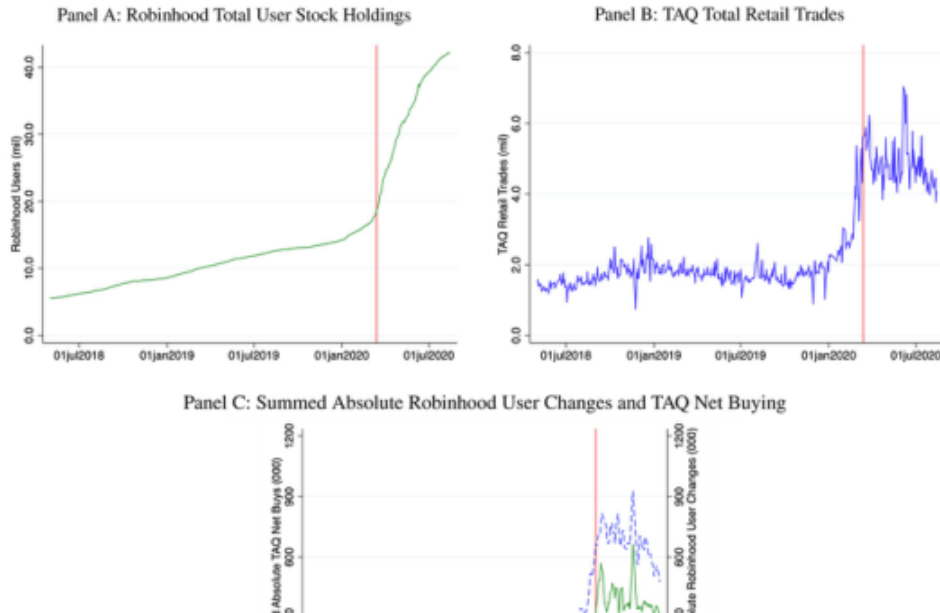


Figure 1: Growth in Robinhood user-stock holdings and retail trading activity.

Read: Robinhood user-stock positions grow sharply over the sample, especially after the COVID-19 national emergency declaration. TAQ retail trading also rises strongly during the same period.

Economic message: The paper studies a period in which Robinhood became large enough and active enough to matter for retail trading, especially in attention-grabbing small stocks.

5.2 Table III: Concentration of retail buying and selling

Table III
Concentration of Retail Buying and Selling

The sample consists of stocks with a measure of Robinhood user changes and net retail buying in TAQ. Statistics are based on averages across days. In Panel A, *HH_buy* is the Herfindahl-Hirschman index for stocks with net buying (sum of squared share of buying) and *Top10_buy* is the percentage of all net buying in the 10 stocks with the highest level of net buying. In Panel B, *HH_sell* is the Herfindahl-Hirschman index for stocks with net selling (sum of squared share of selling) and *Top10_sell* is the percentage of all net selling in the 10 stocks with the highest level of net selling. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Robinhood User Changes	TAQ Retail Trades	Difference (RH – TAQ)
Panel A: Mean Daily Concentration Measures among Stocks with Net Buying			
<i>HH_buy</i> (bps)	246.952*** (13.091)	110.820*** (5.575)	136.133*** (11.433)
<i>Top10_buy</i> (%)	34.621*** (0.412)	23.560*** (0.315)	11.061*** (0.361)
Panel B: Mean Daily Concentration Measures among Stocks with Net Selling			
<i>HH_sell</i> (bps)	172.760*** (20.069)	48.566*** (2.357)	124.194*** (20.055)
<i>Top10_sell</i> (%)	25.370*** (0.369)	14.076*** (0.218)	11.294*** (0.384)

(from trades). The correlation between these two series is 87%: they gener-

Figure 2: Concentration of Robinhood user changes versus TAQ retail trades.

Read: The top 10 stocks account for about 34.6% of Robinhood net buying versus 23.6% of TAQ retail net buying. Robinhood selling is also more concentrated than TAQ retail selling, but buying concentration is the central attention-based prediction.

Economic message: Robinhood users are more coordinated than the broader retail population. This is consistent with many users being pushed toward the same attention-grabbing stocks.

5.3 Table IV: Determinants of Robinhood herding

Table IV
Determinants of Robinhood Herding Indicator Variable
(Top 0.5% of Positive Percentage User Change)

The table examines the determinants that predict the Robinhood herding indicator variable (top 0.5% of positive user change ratio on day t and a minimum of 100 users on day $t - 1$) using a linear probability model. $rh_herd(t - 1)$ is the lagged Robinhood herding indicator. Extreme absolute return ($t - 1$) is an indicator that equals one if the absolute return is ranked in the top 20 on day $t - 1$. Abnormal $Vol(t - 1)$ is the logarithm of the ratio of stock market volume on day $t - 1$ to the average volume from days $t - 21$ to $t - 2$. User Change ($t - 1$) is the change in Robinhood users from day $t - 2$ to day $t - 1$. $ln(Users(t - 1))$ is the logarithm of Robinhood users before the market closes on day $t - 1$. Abnormal $SVI(t - 1)$ is the logarithm of the ratio of Google search volume on day $t - 1$ to the average Google search volume from days $t - 21$ to $t - 2$. Abnormal News ($t - 1$) is the logarithm of the ratio of news article count on day $t - 1$ to the average news article count from days $t - 21$ to $t - 2$. Earnings Announcement ($t - 1$) is an indicator that equals one if the firm has an earnings announcement on day $t - 1$. Robust standard errors are clustered by day and stock level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	$rh_herd(t)$			
	(1)	(2)	(3)	(4)
$rh_herd(t - 1)$	0.106*** (0.005)	0.103*** (0.005)	0.102*** (0.005)	0.102*** (0.005)
Extreme absolute return ($t - 1$) =1: top 20 absret; =0: otherwise	0.0505*** (0.003)	0.0494*** (0.003)	0.0485*** (0.003)	0.0486*** (0.003)
Abnormal $Vol(t - 1)$ = $ln(Vol(t - 1)/AvgVol(t - 21, t - 2))$	0.000551*** (0.000)	0.000453*** (0.000)	0.000368*** (0.000)	0.000348*** (0.000)
User change ($t - 1$) (in 000s)		0.00806*** (0.001)	0.00738*** (0.001)	0.00733*** (0.001)
$ln(Users(t - 1))$		0.000193*** (0.000)	0.000170*** (0.000)	0.000166*** (0.000)
Abnormal $SVI(t - 1)$ = $ln(SVI(t - 1)/AvgSVI(t - 21, t - 2))$			0.000149*** (0.000)	0.000142*** (0.000)
Abnormal news ($t - 1$) = $ln(News(t - 1)/AvgNews(t - 21, t - 2))$			0.00195*** (0.000)	0.00107*** (0.000)
Earnings announcement ($t - 1$)				0.00875*** (0.001)
Observations	3,792,584	3,792,584	3,792,584	3,792,584
R^2	0.022	0.022	0.023	0.023

computed as the logarithm of the ratio of the value on day $t - 1$ to the average from day $t - 21$ to $t - 2$. The lagged herding indicator is also included in

Figure 3: Attention proxies predicting Robinhood herding events.

Read: Lagged herding, top-20 absolute returns, abnormal volume, lagged user changes, Robinhood popularity, abnormal Google search, abnormal news, and earnings announcements all positively predict Robinhood herding.

Economic message: The extreme user-count jumps are not random. They occur in stocks that are especially visible to investors.

5.4 Table V: Robinhood outages and percent retail trade

Table V
The Effect of Robinhood Outages on Percent Retail Trade

The dependent variable is the proportion of TAQ trades that are identified as retail trades per period. *Outage* is an indicator variable that takes a value of one at the time of an outage. In Panels B and C, *Repair* is an indicator variable that takes a value of one for the hour after systems are fully operational. In Panel C, *Partial* is an indicator variable that takes a value of one for the period in which systems are partially restored. Column (1) presents results for all stocks; column (2) for the 50 most popular Robinhood stocks, and column (3) for the 50 high-attention stocks (based on fitted values of model 3, Table IV). In Panel A, the Robinhood outage is for the full day on March 2, 2020, and observations are stock-hours. In Panel B, the Robinhood outage starts March 3, 2020 at 10:15 am, with all systems back online sometime between 11 am and noon; observations are stock-hours. The data set for Panels A and B spans February 18 to March 17. In Panel C, the Robinhood outage starts June 18, 2020, at 10:39 am, with systems improvement at 12:43 pm and fully restored at 1:08 pm. The data set for Panel C spans June 4 to July 2. *Outage* is an indicator variable that takes a value of one for the time intervals between 11:35 am and 12:35 pm, June 18; observations are stock-five-minute periods. Robust standard errors are double clustered by day and ticker. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	All Stocks	50 Popular Stocks	50 High Attention Stocks
Panel A: March 2 Outage (All Day)			
<i>Outage</i>	-0.723*** (0.203)	-5.227*** (0.338)	-4.948*** (0.340)
Observations	1,090,382	8,395	8,319
R^2	0.393	0.745	0.618
Day FE	NO	NO	NO
Ticker FE	YES	YES	YES
Time of Day FE	YES	YES	YES
Panel B: March 3 Outage (Late Morning)			
<i>Outage</i>	-1.720*** (0.172)	-6.610*** (0.712)	-5.122*** (0.550)
<i>Repair</i>	1.581*** (0.133)	3.742*** (0.131)	1.652*** (0.124)
Observations	1,038,326	7,997	7,821
R^2	0.397	0.764	0.614
Day FE	YES	YES	YES
Ticker FE	YES	YES	YES
Time of Day FE	YES	YES	YES
Panel C: June 18 Outage (Late Morning)			
<i>Outage</i>	-0.678*** (0.067)	-3.042*** (0.455)	-0.752** (0.350)
<i>Partial</i>	0.476*** (0.091)	1.005*** (0.312)	0.973** (0.346)
<i>Repair</i>	0.731*** (0.092)	1.543*** (0.190)	0.458*** (0.145)
Observations	9,443,439	81,814	64,652
R^2	0.231	0.621	0.176
Day FE	YES	YES	YES
Ticker FE	YES	YES	YES
Time of Day FE	YES	YES	YES

Figure 4: Retail trading falls during Robinhood outages.

Read: During outages, the retail share of trading falls. The decline is especially large among popular Robinhood stocks and high-attention stocks. Retail trading rises again in repair periods.

Economic message: Robinhood users account for a meaningful part of retail trading in attention-grabbing stocks. Outages provide direct evidence that their trading is not merely correlated with retail activity; it helps generate it.

5.5 Figure 4 and Table VI: Top Movers and gainers versus losers

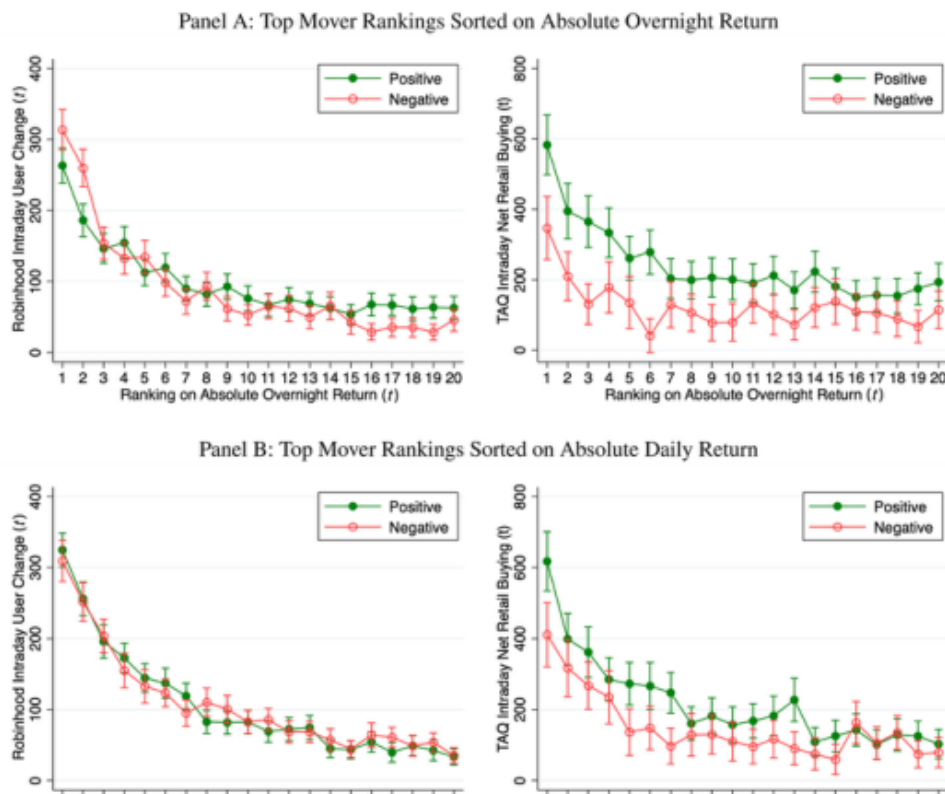


Figure 5: Robinhood and TAQ buying around Top Movers rankings.

Table VI

The Effect of Top Movers on Robinhood Intraday User Change versus TAQ Intraday Net Retail Buying

The table examines how the top mover rankings affect Robinhood (RH) intraday user change and TAQ intraday net retail buying. The rankings are sorted for stocks with market cap above \$300 million at the market open of day t . The stocks are sorted on absolute overnight return (columns (1) and (2)) and absolute daily return (columns (3) and (4)), respectively. The sample requires both Robinhood user change and TAQ net retail buying be available and only includes the top 20 stocks for each day. RH intraday user change measures the change in RH users from first time stamp that excludes market open trades to the last time stamp before the market closes. TAQ Intraday Net Retail Buying is TAQ retail buying minus TAQ retail selling (with short trades removed following Boehmer and Song (2020)) during market trading hours. Top mover score assigns a score for each rank, with 20 for the highest absolute return and 1 for the 20th highest. Negative return is an indicator variable that equals one if the stock return is negative. Regressions include day fixed effects. Robust standard errors are clustered by day. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Top Mover Score is Sorted on:	Absolute Overnight Return (t)		Absolute Return (t)	
	RH Intraday User Change (t)	TAQ Intraday Net Buy (t)	RH Intraday User Change (t)	TAQ Intraday Net Buy (t)
	(1)	(2)	(3)	(4)
Top mover score	7.198*** (0.510)	13.41*** (1.701)	11.95*** (0.539)	17.89*** (1.671)
=20: highest absret; =1: 20 th highest absret				
Negative return	46.49*** (9.913)	-133.2*** (29.950)	2.980 (10.252)	-86.90*** (28.705)
=1: top mover return is negative; =0, otherwise				
Top mover score $\times$ Negative return	3.462*** (0.736)	-6.591*** (2.369)	-1.472** (0.724)	-6.245*** (2.222)
Day FE	Yes	Yes	Yes	Yes
Observations	10,247	10,247	10,342	10,342
R^2	0.233	0.211	0.272	0.192

Figure 6: Regression evidence on Top Movers and Robinhood versus TAQ buying.

Read: Robinhood buying responds strongly to Top Movers ranks for both positive and negative return stocks. TAQ retail buying responds much more favorably to gainers than losers.

Economic message: The way information is displayed matters. By mixing gainers and losers, Robinhood appears to focus attention on extreme price movement itself, not just on winners.

5.6 Table VII: Regression discontinuity around the Top Movers market-cap cutoff

Table VII
Regression Discontinuity in Robinhood Intraday User Change for Top Mover Stocks

This table estimates a sharp regression discontinuity regression that exploits the discontinuity in Robinhood (RH) intraday user changes around the market cap cutoff of \$300 million. Specifically, we use RH intraday user changes at day t as the dependent variable, and include an indicator that equals one for stocks that have market cap at the market open of day t greater than \$300 million. We include different polynomial functions of market cap as controls. Our analysis varies sample bandwidth from Panels A to D. Panel A uses a sample bandwidth of \$50 million (i.e., market cap $\in$ [\$250 million, \$350 million]). For stocks with market cap above \$300 million, we select stocks that rank top 20 by absolute day- t overnight returns among all stocks with market cap above \$300 million. For stocks with market cap below \$300 million, we include matched stocks with absolute day- t overnight returns close to stocks with market cap above \$300 million. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	RH Intraday User Change		
	(1)	(2)	(3)
Panel A: Sample Bandwidth = 50 m			
Larger than 300 m	95.71** (39.472)	115.8** (53.933)	247.3*** (78.782)
Polynomial order, N	1	2	3
Observations	1,332	1,332	1,332
R^2	0.013	0.014	0.020
Panel B: Sample Bandwidth = 75 m			
Larger than 300 m	90.57** (37.123)	109.8* (61.131)	97.18 (80.490)
Polynomial order, N	1	2	3
Observations	2,068	2,068	2,068
R^2	0.006	0.006	0.008
Panel C: Sample Bandwidth = 100 m			
Larger than 300 m	77.55** (33.300)	131.0** (53.648)	174.8** (72.724)
Polynomial order, N	1	2	3
Observations	2,782	2,782	2,782
R^2	0.011	0.011	0.012
Panel D: Sample Bandwidth = 125 m			
Larger than 300 m	49.13* (26.718)	102.3** (41.947)	147.7** (59.496)
Polynomial order, N	1	2	3
Observations	3,406	3,406	3,406
R^2	0.016	0.016	0.016

Figure 7: Regression discontinuity evidence from the \$300 million Top Movers threshold.

Read: Stocks just above the \$300 million cutoff receive significantly more Robinhood intraday user changes than matched stocks just below the cutoff, across several bandwidths and polynomial specifications.

Economic message: This is strong evidence that Robinhood's interface directly affects buying.

Display eligibility creates a discontinuous jump in user behavior.

5.7 Table VIII and Figure 6: Returns around herding events

Table VIII
Event-Time Abnormal Returns

The table reports abnormal returns around Robinhood user herding events. Abnormal returns (*AR*) are computed as the raw return minus the CRSP value-weighted average return. Abnormal returns are averaged across all events. Buy-and-hold abnormal returns (*BHAR*) are computed as the product of one plus the stock's return through event day t less the product of one plus the market return for the same period. Standard errors are computed by clustering on event day. *%Positive* is the percent of returns that are positive. Herding events are defined as the top 0.5% of stocks with positive user change ratio on day 0 and a minimum of 100 users on the prior day. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Event Day	AR	Std. Err.	% Positive	BHAR	Std. Err.	% Positive
Preevent						
-10	-0.08%	0.14%	47%	-0.08%	0.14%	47%
-9	-0.05%	0.15%	46%	-0.02%	0.27%	46%
-8	0.38%	0.30%	47%	0.22%	0.41%	46%
-7	-0.06%	0.21%	45%	-0.01%	0.50%	45%
-6	-0.28%**	0.13%	46%	-0.48%	0.42%	44%
-5	0.02%	0.15%	48%	-0.47%	0.45%	45%
-4	0.39%*	0.20%	48%	-0.10%	0.50%	46%
-3	0.51%**	0.21%	48%	0.52%	0.59%	47%
-2	0.40%*	0.22%	49%	0.81%	0.60%	48%
-1	4.59%***	0.52%	56%	5.60%***	0.87%	51%
0	13.95%***	0.92%	63%	22.16%***	1.73%	58%
Postevent						
1	-1.23%***	0.24%	42%	-1.23%***	0.24%	42%
2	-0.85%***	0.18%	42%	-2.11%***	0.31%	40%
3	-0.43%***	0.16%	44%	-2.74%***	0.29%	38%
4	-0.35%**	0.15%	43%	-3.15%***	0.31%	37%
5	-0.32%**	0.14%	44%	-3.55%***	0.32%	37%
6	-0.37%***	0.13%	44%	-3.99%***	0.34%	37%
7	-0.14%	0.14%	46%	-4.17%***	0.37%	36%
8	0.17%	0.17%	45%	-4.13%***	0.39%	36%
9	0.07%	0.13%	45%	-4.19%***	0.40%	36%
10	0.15%	0.16%	45%	-4.15%***	0.40%	37%
11	-0.03%	0.14%	42%	-4.33%***	0.40%	36%
12	-0.03%	0.12%	46%	-4.42%***	0.41%	36%
13	-0.06%	0.13%	44%	-4.51%***	0.43%	36%
14	-0.09%	0.14%	45%	-4.68%***	0.44%	35%
15	-0.15%	0.10%	45%	-4.87%***	0.45%	35%
16	0.03%	0.11%	46%	-4.83%***	0.47%	35%
17	-0.04%	0.14%	45%	-4.76%***	0.56%	35%
18	0.18%	0.22%	46%	-4.71%***	0.56%	35%
19	-0.01%	0.11%	46%	-4.80%***	0.57%	35%
20	0.15%	0.16%	46%	-4.74%***	0.58%	35%

Figure 8: Event-time abnormal returns around Robinhood herding.

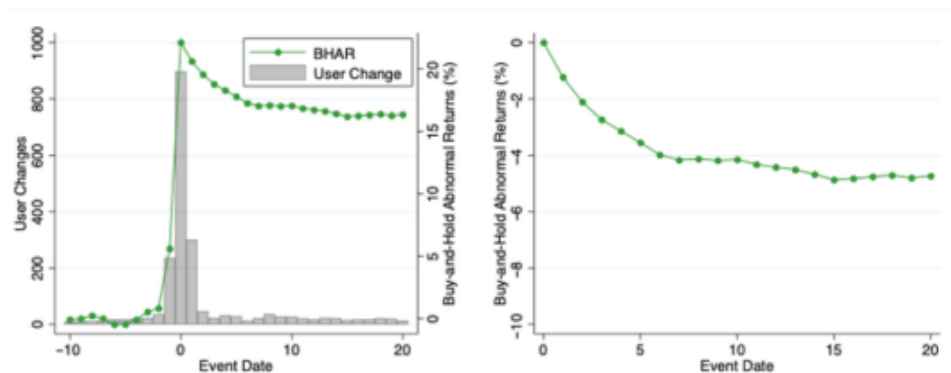


Figure 6. Returns around herding events. The panel on the left depicts mean BHARs and Robinhood user changes from 10 days before to 21 days after herding events. The green line represents BHARs, whereas the gray bars represent user changes. The panel on the right displays

Figure 9: Buy-and-hold abnormal returns and user changes around herding events.

Read: The herding day itself has a very large positive abnormal return, about 14%. Afterward, returns reverse sharply: the 20-day BHAR is about -4.7% .

Economic message: Robinhood users appear to buy after attention-driven price pressure. The subsequent reversal is large, fast, and economically meaningful.

5.8 Figure 7: Herding intensity and return reversal

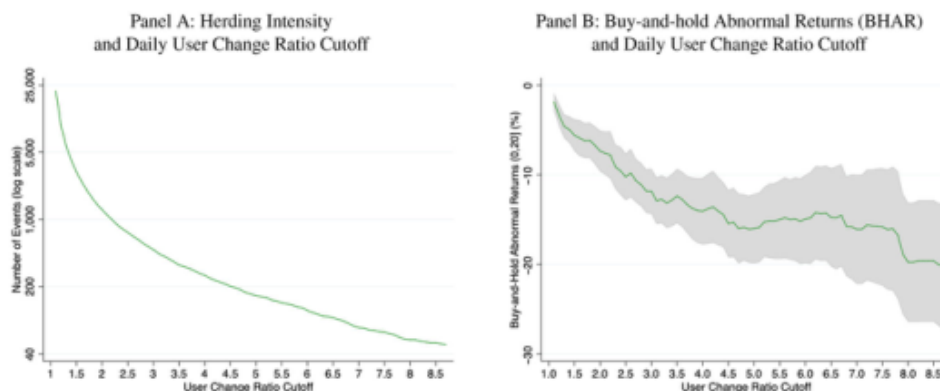


Figure 7. Herding intensity and price reversal. The panels show how herding intensity and 20-day BHARs vary with the daily user change ratio cutoff to identify herding episodes for stocks with at least 100 Robinhood users. Panel A plots the number of herding episodes (in log scale)

Figure 10: More intense herding predicts more negative subsequent returns.

Read: As the threshold for user-count increases becomes more extreme, the number of events falls but the subsequent 20-day BHAR becomes much more negative, reaching roughly -20% for the most extreme events.

Economic message: The return effect scales with the intensity of coordinated buying. This supports a price-pressure interpretation.

5.9 Tables IX–XI: Calendar-time and regression return tests

Read:

- The calendar-time strategy that buys herding stocks earns large negative daily alphas, roughly -55 to -61 basis points over the full period and even more negative alphas during the COVID period.
- Daily stock-return regressions show that lagged Robinhood herding predicts negative returns over the next five days.
- The five-day return effect remains negative after controlling for lagged returns and TAQ retail order imbalance.
- TAQ herding also predicts negative returns, but Robinhood herding is a stronger signal. The most negative returns occur when Robinhood and TAQ herding overlap.

Economic message: The negative-return result is not an artifact of one event-study method. It appears in portfolio tests, panel regressions, and comparisons with general retail herding.

5.10 Figure 8: Aggregate Robinhood investor experience

Read: The paper computes the average purchase and sale experience of the Robinhood community around herding episodes. On average, Robinhood users lose money during these episodes, both in raw terms and relative to a counterfactual S&P 500 ETF trade.

Economic message: The return reversal is not merely a statistical pattern. It maps into poor realized outcomes for the investors who collectively buy during herding events.

5.11 Figure 9 and Table XIII: Short trading around Robinhood herding

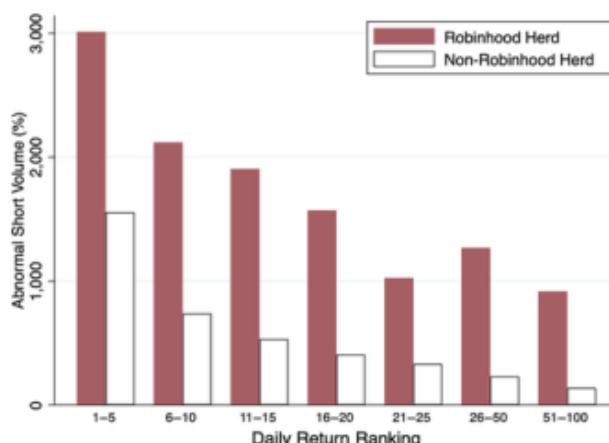


Figure 11: Abnormal short volume is especially high for Robinhood herding stocks.

Table XIII
Regression of Abnormal Short Trading on Returns and Top Robinhood Changes

The dependent variable is abnormal short volume. Short volume is measured using data from FINRA, which provides daily short trade data as noted by Boehmer and Song (2020). We compute abnormal short volume on day t as the ratio of short trades on day t to the prior 20-day ($t - 20, t - 1$) average of the number of short trades. The top 0.1% of observations are winsorized. The key independent variables are *rh_chgratio*, which is the percentage change in users from day $t - 1$ to t , and *rh_herd*, which is an indicator variable that equals one if the percentage change in users is in the top 0.5% for stocks with positive user changes on day t and a minimum of 100 users on day $t - 1$. Control variables include the excess return of day t , dummy variables representing the return ranks (*ret_rank*) of day t , and abnormal news coverage, which is the logarithm of the ratio of news article count on day t to the average news article count from day $t - 20$ to $t - 1$. Coefficients are in percent. Standard errors are computed using Fama-Macbeth (1973). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Abnormal Short Volume (%)			
	(1)	(2)	(3)	(4)
<i>rh_chgratio</i>	637.14*** (12.75)		551.10*** (11.10)	
<i>rh_herd</i>		854.65*** (14.5)		686.57*** (17.25)
Excess return	499.54*** (16.32)	498.83*** (15.95)		
Abnormal News	6.64*** (0.13)	7.17*** (0.14)	6.37*** (0.12)	6.85*** (0.13)
<i>Ret_rank_1</i>			858.14*** (47.95)	1049.44*** (36.36)
<i>Ret_rank_2</i>			704.80*** (44.31)	818.70*** (34.22)
<i>Ret_rank_3</i>			568.94*** (68.35)	670.82*** (32.64)
<i>Ret_rank_4</i>			583.67*** (35.00)	635.83*** (31.91)
<i>Ret_rank_5</i>			534.46*** (30.95)	554.07*** (30.50)
<i>Ret_rank_6</i>			465.52*** (26.90)	480.40*** (26.95)
<i>Ret_rank_7</i>			410.78*** (39.91)	482.44*** (28.08)
<i>Ret_rank_8</i>			371.97*** (27.60)	398.30*** (25.01)
<i>Ret_rank_9</i>			381.49*** (26.37)	395.11*** (26.27)
<i>Ret_rank_10</i>			361.47*** (24.36)	385.87*** (24.53)
<i>Ret_rank_11 – 25</i>			269.46*** (8.20)	279.53*** (8.55)
<i>Ret_rank_26 – 50</i>			145.48*** (4.30)	152.75*** (4.56)
<i>Ret_rank_51 – 100</i>			83.45***	86.87***

Figure 12: Regression evidence on abnormal short trading and Robinhood herding.

Read: Abnormal short volume is much higher for stocks experiencing Robinhood herding. In regressions, both the user-change ratio and the herding indicator strongly predict abnormal short volume, even after controlling for returns, return ranks, and abnormal news.

Economic message: Other market participants appear to trade against Robinhood herding. This suggests that the predictable reversals are visible and economically exploitable.

6 Short summary

- The paper studies whether Robinhood’s fintech trading environment induces attention-based retail trading and whether this trading affects returns.
- The key data source is Robintrack, which records how many Robinhood users hold each security over time.
- Robinhood users exhibit more concentrated buying than broader retail investors.
- Attention variables predict extreme Robinhood purchase herding.
- Robinhood outages reduce retail trading especially in Robinhood-popular and high-attention stocks.
- Robinhood’s Top Movers list affects trading: users buy both extreme gainers and extreme losers, and stocks just above the \$300 million display cutoff receive extra buying.
- Intense Robinhood buying predicts large negative returns over the next several days and weeks.
- The negative-return effect is stronger for more extreme herding, small stocks, and the high-retail-trading COVID period.
- Short sellers increase activity around Robinhood herding events, consistent with trading against predictable attention-induced price pressure.

7 Conclusion

7.1 Core conclusion

The paper shows that Robinhood users engage in unusually concentrated attention-induced trading. This behavior is not explained solely by general retail trading or by stock fundamentals. It is partly shaped by the app’s information display, especially the Top Movers list.

The main return implication is that extreme Robinhood purchase herding predicts negative subsequent abnormal returns. The baseline herding stocks lose around 5% over the next month, and the most intense events lose much more.

7.2 Economic intuition

Robinhood reduces trading frictions and simplifies the investment interface. This can increase market access, but it also makes it easier for many inexperienced or speculative investors to react to the same salient signals. When many investors buy the same attention-grabbing stocks at the same time, their trades can temporarily push prices away from fundamentals. Prices then reverse as attention fades, liquidity suppliers respond, or short sellers trade against the pressure.

7.3 Takeaway

The broader lesson is that disclosure and access are not enough. **How information is displayed** can meaningfully shape investor decisions. A simple app can help investors participate in markets, but it can also coordinate attention in ways that generate herding, temporary price pressure, predictable reversals, and losses for the investors drawn into the episodes.